BIOS 735: Shrinkage Regression Methods

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"There's two buttons I never like to hit: that's panic and snooze"

Introduction

- Today, we'll talk about shrinkage methods
- All shrinkage methods are biased (by design) even when the model is correct.
- The will, however, result in lower variance than the full LR model.
- Some shrinkage methods we'll discuss:
  - Ridge Regression
  - Lasso
  - Elastic Net
  - Least Angle Regression (LAR)

Instability of LS Estimators

- Recall that $\hat{\beta} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$ can be unstable
- Further, when $rank(\boldsymbol{X}) < p$ $\boldsymbol{X}'\boldsymbol{X}$ is singular
  - $\boldsymbol{X}$ is ill-conditioned
  - columns of $\boldsymbol{X}$ are collinear
  - $p > n$
- We can measure ill-conditioning by $\kappa = d_1/d_p$ where d_1 and d_p are the largest and smallest eigenvalues, respectively, of $\boldsymbol{X}$.
- We can measure collinearity for a predictor X_k via $VIF_k = (1 - R_k^2)^{-1}$ where R_k^2 is the R^2 for all other covariates regressed on X_k

Biased Regression Methods

- The instability of the LS estimates is due to $\boldsymbol{X}'\boldsymbol{X}$ being singular or nearly singular.
- **Solution** allow $\hat{\beta}$ to be biased.
- We'll assume that $\boldsymbol{x}$ and $\boldsymbol{y}$ have been centered (no β_0). In practice, we will center and scale these data.

Ridge Regression

- Potential instability in $(\boldsymbol{X}'\boldsymbol{X})^{-1}$ could be solved by adding a constant λ to $\boldsymbol{X}'\boldsymbol{X}$.
- Adding a constant to the diagonal of $\boldsymbol{X}'\boldsymbol{X}$ will bring it closer to a matrix with 0's on the off diagonal.
- This results in the ridge regression estimate of $\hat{\beta}$

$$\hat{\beta}^{\text{ridge}} = (\boldsymbol{X}'\boldsymbol{X} + \lambda \boldsymbol{I})^{-1}\boldsymbol{X}'\boldsymbol{y}$$

where $\boldsymbol{I}$ is the $p \times p$ identity matrix.

Ridge Regression Properties

- Another way to view the ridge regression model is that

$$\hat{\boldsymbol{\beta}}^{\text{ridge}} = \text{argmin}_{\beta} \left\{ (\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta}) + \lambda ||\boldsymbol{\beta}||^2 \right\},$$

  where $||\boldsymbol{\beta}||^2 = \sum_{j=1}^{p} \beta_j^2$ (note: the λ here is the same as on the previous slide)
- (Yet) Another way to view the ridge regression model is that $\hat{\boldsymbol{\beta}}^{\text{ridge}}$ minimizes

$$ESS(\boldsymbol{\beta}) = (\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta}) \quad \text{such that} \quad ||\boldsymbol{\beta}||^2 \leq t,$$

  which makes explicit the size constraint on the parameters.
- The relationship between t and λ is made through Lagrange multipliers.

Ridge Regression Properties

- The bias and variance of Ridge coefficients have nice closed forms

$$
\begin{aligned}
Var_\lambda(\hat{\beta}^{\text{ridge}}) &= \sigma^2 \sum_{j=1}^{p} \frac{d_j^2}{(d_j^2 + \lambda)^2} \\
Bias_\lambda^2(\hat{\beta}^{\text{ridge}}) &= \lambda^2 \sum_{j=1}^{p} \frac{(\hat{\alpha}_j^{\text{ridge}})^2}{(d_j^2 + \lambda)^2}
\end{aligned}
$$

thus,

$$
MSE_\lambda(\hat{\beta}^{\text{ridge}}) = \sigma^2 \sum_{j=1}^{p} \frac{\sigma^2 d_j^2 + \lambda(\hat{\alpha}_j^{\text{ridge}})^2}{(d_j^2 + \lambda)^2}
$$

Lasso regression model

- The Lasso regression model, results in solution $\hat{\beta}^{\text{lasso}}$ minimizes

$$ESS(\boldsymbol{\beta}) = (\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta}) \quad \text{such that} \quad \sum_{i=1}^{p} |\beta_j| \leq t,$$

which makes explicit the size constraint on the parameters.

- The solution to this is such that

$$\hat{\boldsymbol{\beta}}^{\text{lasso}} = \text{argmin}_{\beta} \left\{ (\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta}) + \lambda \sum_{i=1}^{p} |\beta_j| \right\},$$

- The relationship between these forms is made through Lagrange multipliers.

Lasso properties

- A major difference between the Lasso and Ridge models is that making t sufficiently small (or λ large) will result in some β coefficients being exactly zero.
- Let $t_0 = \sum_{i=1}^{p} |\hat{\beta}_j^{ls}|$ then if $t = t_0/2$ the Lasso will result in coefficients that are the least squares coefficients shrunk by about 50% on average.
- There is no closed-form solution for Lasso coefficients (they are nonlinear in $\boldsymbol{y}$).

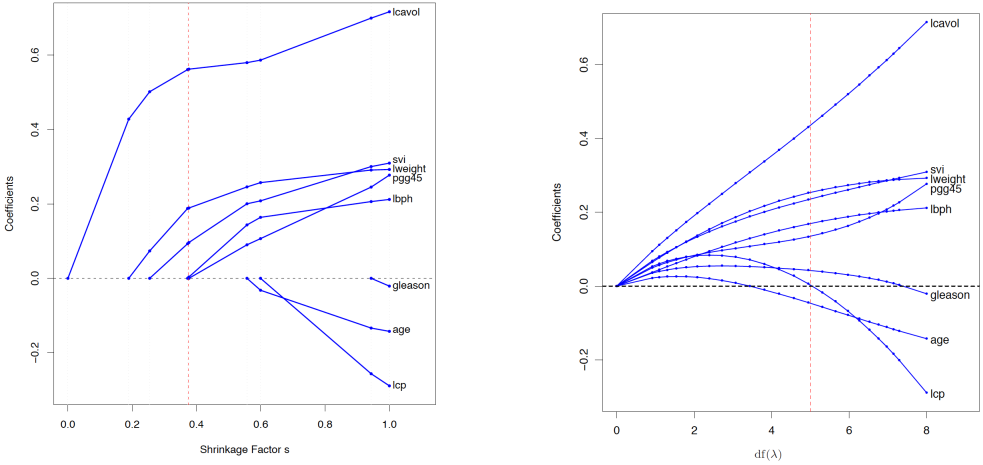


Figure: From ESL, where $s = t / \sum_1^p |\hat{\beta}_j^{ls}|$

Ridge vs Lasso vs Elastic Net

- The Lasso regression model,

$$\hat{\boldsymbol{\beta}}^{\text{lasso}} = \text{argmin}_{\beta} \left\{ (\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta}) + \lambda \sum_{i=1}^{p} |\beta_j| \right\},$$

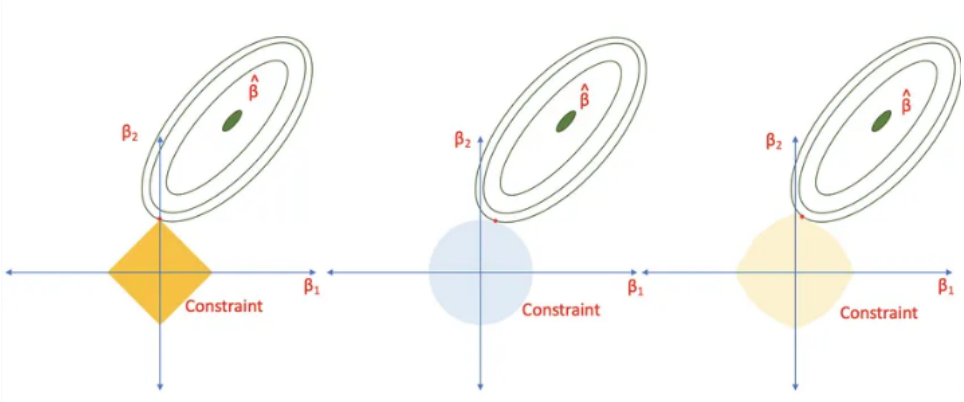
- The ridge regression model,

$$\hat{\boldsymbol{\beta}}^{\text{ridge}} = \text{argmin}_{\beta} \left\{ (\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta}) + \lambda \sum_{i=1}^{p} \beta_j^2 \right\},$$

- The elastic net regression model (Zou and Hastie, 2005),

$$\hat{\boldsymbol{\beta}}^{\text{en}} = \text{argmin}_{\beta} \left[(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta}) + \lambda \sum_{i=1}^{p} \{\alpha\beta_j^2 + (1-\alpha)|\beta_j|\} \right],$$

Ridge vs Lasso vs Elastic Net



Ridge vs Lasso vs Best Subset

- Suppose $\boldsymbol{X}$ is a standardized orthonormal input matrix
- The estimators for Ridge, Lasso, and best subset, denoted by $\hat{\beta}^{\text{ridge}}$, $\hat{\beta}^{\text{lasso}}$ and $\hat{\beta}^{\text{bss}}$ respectively, are

$$
\begin{aligned}
\hat{\beta}_j^{\text{bss}} &= \hat{\beta}_j^{ls} \times I\left(|\hat{\beta}_j^{ls}| \geq |\hat{\beta}_{(M)}|\right) \text{ of size M} \\
\hat{\beta}_j^{\text{ridge}} &= \frac{\hat{\beta}_j^{ls}}{1+\lambda} \\
\hat{\beta}_j^{\text{lasso}} &= \text{sign}\left(\hat{\beta}_j^{ls}\right)\left(|\hat{\beta}_j^{ls}| - \lambda\right)_+
\end{aligned}
$$

where $\hat{\beta}_{(M)}$ is the Mth largest regression coefficient.

Other penalties

- Last class, we went over Ridge, LASSO, and elastic net regression models.
- The optimization functions were all of the form

$$\text{argmin}_{\beta} \left\{ (\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta}) + p(\boldsymbol{\beta}; \lambda) \right\},$$

  where $p(\boldsymbol{\beta}; \lambda)$ is a penalty function.
- There are other options for $p(\boldsymbol{\beta}; \lambda)$.

Minimax Concave Penalty (MCP)

- Given a tuning parameter $\lambda > 0$ and a parameter $a > 1$, the MCP penalty for a coefficient β is defined as:

$$p_{MCP}(\beta; \lambda, a) = \begin{cases} \lambda|\beta| - \frac{\beta^2}{2a} & \text{if } |\beta| \leq a\lambda \\ \frac{a\lambda^2}{2} & \text{if } |\beta| > a\lambda \end{cases}$$

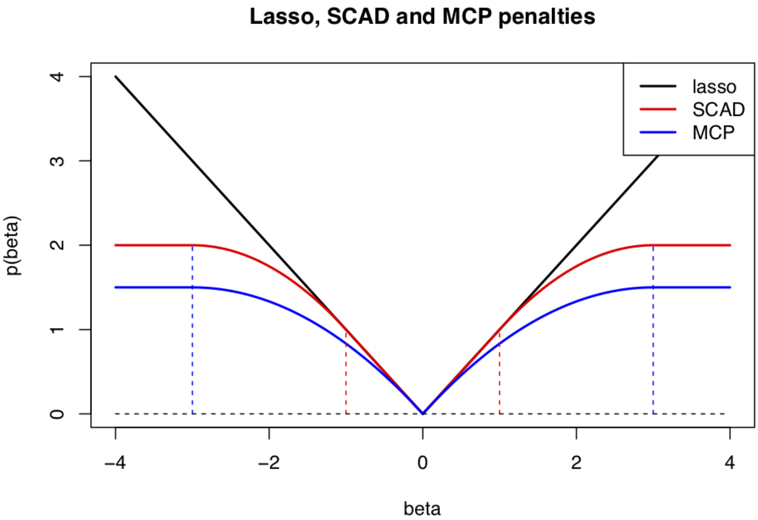
- The first segment is concave, and the penalty becomes constant after $|\beta| > a\lambda$.
- The parameter a controls the concavity and thus influences the amount of regularization applied to the coefficients.

Smoothly Clipped Absolute Deviation (SCAD)

- Given a tuning parameter $\lambda > 0$ and a parameter $a > 2$, the SCAD penalty for a coefficient β is defined as:

$$p_{SCAD}(\beta; \lambda, a) = \begin{cases} \lambda|\beta| & \text{if } |\beta| \leq \lambda \\ -\left(\frac{\beta^2 - 2a\lambda|\beta| + \lambda^2}{2(a-1)}\right) & \text{if } \lambda < |\beta| \leq a\lambda \\ \frac{(a+1)\lambda^2}{2} & \text{if } |\beta| > a\lambda \end{cases}$$

- The SCAD penalty starts linearly (similar to the L1 penalty) but then bends down and becomes flat after $|\beta| > a\lambda$.
- The SCAD penalty is continuous everywhere, making it smoother than the MCP.


Lasso, SCAD and MCP penalties
lasso
SCAD
MCP
p(beta)
beta
0
1
2
3
4
−4
−2
0
2
4

Partial Least Squares

- PCR was performed by making a linear combination of the $\boldsymbol{X}$'s, then performing LR on the derived linear combinations.
- Partial Least Squares (PLS) has similarities with Principal Components Analysis (PCA), except that the linear combination of the $\boldsymbol{X}$'s is made using the outcome.

Partial Least Squares Algorithm

1. Standardize each $\boldsymbol{x}_j$ and set $\hat{\boldsymbol{y}}^{(0)} = \boldsymbol{y} - \bar{\boldsymbol{y}}\mathbf{1}$, and $\boldsymbol{x}_j^{(0)} = \boldsymbol{x}_j$, $j = 1, 2, \ldots, p$.
2. For $m = 1, 2, \ldots, M \leq p$
   2.1 $z_{im} = \sum_{j=1}^{p} \hat{\psi}_{mj} x_{ij}^{(m-1)}$, where $\hat{\psi}_{mj}$ is the SLR coefficient for $\boldsymbol{x}_j^{(m-1)}$ on $\boldsymbol{y}^{(m-1)}$

   $$\hat{\psi}_{mj} = \frac{\boldsymbol{x}_j^{(m-1)\prime}\boldsymbol{y}}{\boldsymbol{x}_j^{(m-1)\prime}\boldsymbol{x}_j^{(m-1)}} = \frac{\langle \boldsymbol{x}_j^{(m-1)}, \boldsymbol{y}\rangle}{\langle \boldsymbol{x}_j^{(m-1)}, \boldsymbol{x}_j^{(m-1)}\rangle}$$

   2.2 Get the SLR coefficient for $\boldsymbol{z}_m$ on $\boldsymbol{y}^{(m-1)} \Rightarrow \hat{\theta}_m = \langle \boldsymbol{z}_m, \boldsymbol{y}^{(m-1)}\rangle / \langle \boldsymbol{z}_m, \boldsymbol{z}_m\rangle$
   2.3 $\hat{\boldsymbol{y}}^{(m)} = \hat{\boldsymbol{y}}^{(m-1)} - \hat{\theta}_m \boldsymbol{z}_m$.
   2.4 Orthogonalize each $\boldsymbol{x}_j^{(m-1)}$ with respect to $\boldsymbol{z}_m \Rightarrow \boldsymbol{x}_j^{(m)}$
3. Then the PLSR model with m components is

$$\hat{\boldsymbol{y}}(k) = \bar{\boldsymbol{y}}\mathbf{1} + \sum_{m=1}^{k} \hat{\theta}_m \boldsymbol{z}_m.$$

Grouped Lasso regression model

- In some problems, the predictors belong to predefined groups (e.g., genes belonging to a specific biological pathway or dummy-coded categorical variables).
- In this situation, it may be desirable to shrink and select the members of a group together.
- Suppose that the p predictors are divided into L groups, with p_ℓ being the number of variables in group ℓ
- The Grouped Lasso regression model minimizes

$$(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta}) + \lambda \sum_{\ell=1}^{L} \sqrt{p_\ell}||\boldsymbol{\beta}_\ell||_2,$$

where $\boldsymbol{\beta}_\ell$ are the regression coefficients for group ℓ, and $||\cdot||_2$ is the Euclidean norm.

Fused Lasso regression model

- Along with grouped methods, there is the idea of fusion or merging.
- In this situation, it may be desirable to shrink and select the members of a group towards the same regression coefficient.
- Consider the categorical covariates; by assuming the coefficient for one group is zero, you fuse that group with the referent group.
- The idea of fusion is to fuse groups together and not have the fusion be subject to the reference group.
- The Fused Lasso regression proposed by Tibshirani et al (2005) minimizes

$$\hat{\boldsymbol{\beta}} = arg\min\left\{(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{x}\boldsymbol{\beta})\right\}$$

such that

$$\sum_{j=1}^{p} |\beta_j| \leq s_1 \quad \text{and} \quad \sum_{j=2}^{p} |\beta_j - \beta_{j-1}| \leq s_2.$$